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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACTXLA02 - WET PROCESSING

TECHNIQUES FOR TEXTILES

Part A

Multiple Choice Questions

1. State the condition of fabric during drying in the tentering process.

c) Held in the stretched position

2. State the reason why sizing or waxy substances should be removed before singeing.

b) To prevent scorching and burning

3. State the key difference between waterproof and water-repellent fabrics.

c) Waterproof fabrics block all water, while water-repellent fabrics resist light rain

4. State a key drawback of fibre dyeing.

b) Requires early color decision

5. State the reason why curing is applied after hydro-extraction in tumble dyeing.

b) To fix the dye by heat application (e.g., ironing or steam pressing)

6. State the method between winch and jet dyeing that consumes less water, energy, and chemicals.

b) Jet dyeing

7. State the fibres dyed with disperse dyes.

b) Polyester

8. State the process involved in naphthol dyes.

b) Coupling reaction

9. Identify the fabric most suitable for discharge printing.

c) Cotton dyed with reactive dyes

10. Why is a separate engraved roller required for each color in roller printing?

b) To ensure accurate alignment of multiple colors

Part B

Fill in the Blanks Questions

11. **Sodium chlorite is used for bleaching ..... and protein fibers.**

Answer- Cellulosic

12. **Shearing enhances the fabric's ..... by removing lint and uneven surfaces.**

Answer- appearance/texture/durability

13. Shrink resistance treatment is often used to prevent shrinkage during…..

Answer- laundering process (washing, drying)

14. **In pad dyeing, fabric in open width passes through a dye bath followed by ..... rollers.**

Answer- squeeze rollers

15. Acid dyes are…..... soluble.

Answer- water soluble

16. **Basic dyes are ..... charged.**

Answer- positively charged

17. One major disadvantage of DP finish is the release of…..which can raise health concerns.

Answer- formaldehyde

18. **..... gives a fabric a uniform solid shade even when made of different fibers.**

Answer- Union dyeing

19. One limitation is that rotary screen printing is not economical for..... orders due to its high setup costs.

Answer- small scale

20. Duplex printing prints on ..... sides of the fabric.

Answer- both

Part C

Answer in One Word or One Sentence

21. Name the one-word process that enhances both strength and luster of cotton.

Answer – mercerization

22. **Name one natural sizing agent.**

Answer – starch

23. **Name one industry where flame retardant fabrics are crucial.**

Answer – Fire fighting/ Oil and gas/ Military/ Hospitality and public buildings

24. Define direct dyes.

Answer - Direct dyes are water-soluble anionic dyes that have a natural affinity for cellulosic fibers, such as cotton, linen, rayon, and jute.

25. Mention one property of reactive dyes.

Answer - High colourfastness/ Vibrant shades/ Excellent wash fastness

26. State the alternative name for skein dyeing.

Answer – Hank dyeing

27. **Identify the dyeing method that ensures the most uniform dye distribution in wool fibres.**

Answer – stock dyeing

28. **State the reason gentle motion (swinging or stirring) is used during the dip-dye process.**

Answer - To ensure uniform dye distribution and prevent streaking or patchiness**.**

29. State one factor that makes block printing sustainable.

Answer - Use of natural dyes/ Minimal water consumption/durability/reusability/ reduced energy consumption

30. Jig dyeing is mainly used for which type of textiles?

Answer – woven fabrics

Part D

Short Answer Questions

31. State the physical changes cotton fibers undergo during mercerization.

Answer- during mercerization, the cotton fibers, swell, increasing in diameter and becoming rounder.

32. Define desizing and state its importance in textile processing.

Answer- Desizing is the process of removing the sizing agents—like starch or other polymers—that were applied to the warp yarns before weaving.

Key importance of desizing: Improves absorbency, Ensures even dyeing and printing, Enhances fabric handfeel

33. Explain how chlorination helps in wool shrink-resistance.

Answer- wool fabric is treated with chlorine to remove the surface scales from fibers.

34. Define garment dyeing.

Answer- garment dyeing –also called product dyeing- is the technique of colouring fully constructed garments instead of pre-dyed fabric.

35. State any two advantages of vat dyes.

Answer- Excellent colorfastness, Durable colors

36. Differentiate textile printing from dyeing.

Answer- The core difference is that dyeing colors the entire fabric uniformly**,** while printing applies color to specific, localized areas to create a pattern or design.

Part E

Long Answer Questions

37. Discuss at least three purposes of flocking and the benefits each provides to fabric.

Answer- 1. Aesthetic effect- flocking creates unique textures and visual effects
2. softness- the process adds softness and luxurious feel to the fabric
3. insulation- flocking can provide thermal insulation properties

38. Differentiate between natural dyes and synthetic dyes.

Features	**Natural dyes**	**Synthetic dyes**
Source	Natural	Man-made
Cost	Expensive due to labor-intensive production	Much cheaper to produce
Color range	Limited	A vast spectrum of colors
Colorfastness	Moderate to poor	Engineered for superior fastness to washing, light, and rubbing
Environmental impact	Eco-friendly, biodegradable, and non-toxic.	less eco-friendly

39. **Define fabric dyeing and explain its importance as a stage in textile production.**

Answer- fabric dyeing or piece dyeing, is the process of applying color to fully woven or knitted fabrics after their production. This stage is indispensable in textile manufacturing, as it provides desigh flexibility by enabling color decisions later in the work flow. The process typically unfolds in three key steps: pre-treatment (scouring and bleaching), dyeing (immersing the fabric in a dye bath, post treatment (dye fixation).

40. Explain the main components and step-by-step process of screen printing.

Answer-screen printing is a popular printing technique used to apply designs on to the fabric and other materials using a stencil (called a screen). Ink is pushed through the screen's open areas using a squeegee, transferring the design onto the surface below.

<u>Process of screen printing</u>

1. Prepare a stencil of the design on the screen.
2. Place the screen on the fabric or surface.
3. Spread ink on the screen and use a squeegee (a flat tool with a rubber edge) to push ink through the open areas.
4. Remove the screen and let the print dry.
5. Heat-set the ink to fix the design.
6. Repeat for each color with different screens.